Motov - MotorBike rental management system

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Abstract. The rapid growth of digital platforms has transformed the transportation and rental industry. Traditional motorbike rental services often rely on manual processes, making vehicle management, booking, and payment inefficient. This study proposes Motov, a hybrid motorbike rental platform that integrates both company-owned vehicles and partner-owned vehicles into a unified system. The platform provides web and mobile applications for customers, partners, and administrators. Key functionalities include vehicle management, online booking, payment processing, rental tracking, and review management. The proposed system aims to improve vehicle utilization, increase rental accessibility, and provide a convenient user experience. The project is implemented using ReactJS, React Native, ASP.NET Core, and MongoDB.

Keywords: Motorbike Rental · Vehicle Management · Online Booking · Mobile Application · Web Application

1 Introduction

The demand for motorbike rental services has increased significantly due to tourism growth, urban mobility needs, and the popularity of short-term transportation solutions. In Vietnam, many rental businesses still operate through manual processes such as phone calls, social media messaging, and paper-based agreements.

Traditional rental approaches often face several challenges, including inefficient vehicle management, limited booking transparency, and difficulties in tracking rental transactions. Furthermore, customers may experience inconvenience when searching for available vehicles, while rental businesses struggle to maximize vehicle utilization.

Recent studies have investigated vehicle rental systems, fleet management, customer satisfaction, and peer-to-peer vehicle sharing models [3, 2, 5, 4, 6, 7].

This project proposes Motov, a hybrid motorbike rental platform that supports both company-owned and partner-owned vehicles. The system allows customers to search and book vehicles online, partners to register and manage their vehicles, and administrators to oversee platform operations.

The main contributions of this project are:

– Development of a hybrid rental model integrating company-owned and partner-owned vehicles.

– Design of a centralized vehicle management system.
– Implementation of web and mobile applications for customers, partners, and administrators.
– Improvement of booking convenience and vehicle utilization.

2 Related Work

Several studies have investigated vehicle rental, fleet management, customer satisfaction, and car-sharing systems.

Khwaja and Yang [3] examined the relationship between employee engagement, customer satisfaction, and customer retention in the car rental industry. Their findings indicated that service quality significantly influences customer loyalty and business performance. However, the study focused on customer behavior rather than software platform development.

Gonzalez et al. [2] proposed a fleet utilization metric for shared mobility systems. Although the study improved operational evaluation, it did not address booking management or payment integration.

Munzel et al. [5] analyzed different car-sharing business models and demonstrated the viability of peer-to-peer vehicle sharing. However, the research did not provide a practical software implementation.

Liu et al. [4] investigated booking optimization algorithms for online car-sharing systems. Their work improved vehicle allocation efficiency but lacked user management and payment functionalities.

Neifer et al. [6] explored trust-building mechanisms in peer-to-peer car-sharing platforms. The study highlighted the importance of reputation systems but did not provide a complete rental management solution.

Andersen [1] proposed a queueing model for rental scheduling and inventory optimization. However, the study focused on mathematical optimization rather than platform implementation.

Soppert et al. [7] studied the integration of traditional rental and car-sharing services. Their findings showed improved utilization and profitability, but no web or mobile implementation was proposed.

Although previous studies have focused on fleet utilization, scheduling optimization, customer satisfaction, and peer-to-peer sharing, few studies provide a comprehensive platform that integrates booking, payment processing, vehicle management, partner vehicle registration, and administrative monitoring.

Therefore, this project proposes a hybrid motorbike rental platform that combines company-owned and partner-owned vehicles through a centralized web and mobile ecosystem.

3 Method

3.1 System Architecture

The proposed system adopts a three-tier architecture:

- Frontend: ReactJS
- Mobile Application: React Native
- Backend: ASP.NET Core Web API
- Database: MongoDB

3.2 Actors

- Customer
- Partner
- Administrator

3.3 Main Functionalities

Customer:

- Register
- Login
- Search Vehicles
- View Vehicle Details
- Create Booking
- Payment
- Submit Reviews

Partner:

- Register Vehicles
- Update Vehicle Information
- Manage Rental Requests
- View Rental History

Administrator:

- Manage Users
- Manage Vehicles
- Manage Bookings
- Generate Reports

4 System Analysis

4.1 Problem Analysis

Current rental services often rely on manual management processes, resulting in poor booking transparency, inefficient vehicle tracking, and limited operational control.

4.2 Proposed Solution

The proposed solution is a centralized motorbike rental platform supporting both company-owned and partner-owned vehicles through web and mobile applications.

5 System Design

5.1 Use Case Design

Main actors:

- Customer
- Partner
- Administrator

Main use cases:

- Authentication
- Vehicle Management
- Booking Management
- Payment Management
- Review Management

5.2 User Interface Design

- Login Page
- Register Page
- Home Page
- Vehicle Listing Page
- Vehicle Detail Page
- Booking Page
- Admin Dashboard

6 Database Design

The database contains the following collections:

- Users
- Vehicles
- Bookings
- Payments
- Reviews
- Partners

Figure XXX presents the ERD of the proposed system.

7 Implementation

7.1 Technologies

- ReactJS
- React Native
- ASP.NET Core
- MongoDB
- GitHub
- Postman

7.2 Current Progress

- Requirements Analysis
- Literature Review
- ERD Design
- Use Case Design
- UI Design
- Authentication Module (Login/Register)
- CRUD Functions for Users, Vehicles, and Bookings

8 Conclusion

This paper presented Motov, a hybrid motorbike rental platform integrating company-owned and partner-owned vehicles into a unified rental ecosystem. The proposed system supports customers, partners, and administrators through web and mobile applications. Future work includes recommendation systems, real-time vehicle tracking, and advanced analytics.

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